

The Impact of Robotization on Employees' Mental and Physical Health: A Qualitative Study in the Czech and Slovak Republics

VIOLA HUBAČOVÁ PIROVÁ^{1,†}, VLADIMÍRA LIPŠOVÁ^{1,2†,*}, JANA MURZA¹, ZDENĚK MUSIL¹,
VLADIMÍR MUSIL^{3,4}, IVANA LÍBALOVÁ¹

¹National Institute of Public Health, Prague, Czech Republic

²Department of Occupational Medicine, General University Hospital and First Faculty of Medicine, Charles University, Prague, Czech Republic

³Centre of Scientific Information, Third Faculty of Medicine, Charles University, Prague, Czech Republic

⁴Department of Anatomy, Second Faculty of Medicine, Charles University, Prague, Czech Republic

[†]Both authors contributed equally to this work and share first authorship.

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ABSTRACT

Background: *This study aimed to explore the impact of introducing robotic technologies on employees' mental and physical health across three Czech and one Slovak manufacturing company. It sought to understand employees' subjective experiences and perceptions of organizational support during the robotization process.* **Methods:** *A qualitative study was conducted between 2023 and 2025 in four companies. 27 semi-structured interviews with employees (15 women, 12 men) and 19 unstructured interviews with managers (3 women, 16 men) were conducted on-site during working hours. Data were analyzed using thematic analysis with open and axial coding.* **Results:** *Nine thematic areas were identified: general perceptions, change, fear, problems, support, social relations and communication, employee characteristics, physical health, and mental health. Employees generally perceived robotization positively, noting reduced physical workload and less monotonous or hazardous work. Initial adaptation was challenging, especially due to technical demands and workflow changes. Technical staff experienced higher stress than operators, related to responsibility, troubleshooting, and learning new skills. Technical malfunctions, elevated noise and vibration, and cognitive demands were reported as stressors. Personifying robots and informal knowledge sharing facilitated adaptation. No significant sleep disturbances or substance use linked to robotization were reported. Managers faced a high mental workload due to coordination and supervision responsibilities.* **Conclusions:** *When the implementation process is well designed, robotization does not necessarily negatively affect employees' mental health. The findings highlight the importance of actively addressing organizational and human factors that influence the success of technological change in the workplace. Attention should focus on monitoring cognitive and psychological demands, particularly for technical and managerial staff.*

1. INTRODUCTION

The implementation of robotic technologies in the workplace, including both industrial robots and

collaborative robots (cobots), is a key characteristic of the so-called Fourth Industrial Revolution (Industry 4.0), which fundamentally reshapes the organization of production processes, management

systems, and employee requirements [1]. Robotization contributes to increased production efficiency, reduced exposure to risk factors, and fewer workplace injuries [2, 3]. In addition to its effects on physical health, researchers are increasingly focusing on the implications of these changes for employees' psychological well-being [2-4]. The rapid pace of technological change, combined with the need for continuous adaptation, can create uncertainty and stress among workers.

For organizations, robotization represents a profound transformation—often involving not only technological upgrades but also team restructuring, changes in job descriptions, and a shift from routine activities to more technically and mentally demanding tasks [5]. Research shows that such transformations may lead to various forms of psychological stress. Some authors have pointed to elevated stress levels, particularly in connection with robot malfunctions [6], software failures [7], time pressure, and fear of failure [1, 8]. Another important factor is the fear of job loss or demotion, which is especially prevalent among less-skilled workers and may result in frustration or disruption of work identity [9, 10].

On the other hand, some studies [11, 12] highlight potential positive aspects of robotization for mental health, such as reduced workload and monotony, and opportunities for employees to engage in more creative tasks [3], which may enhance job satisfaction and employee engagement. Evidence also suggests [3, 13] that when the introduction of robotic technologies is accompanied by adequate training, accessible technical support, and active communication from management, the negative psychological impacts can be significantly mitigated or even prevented. Furthermore, some research [14, 15] emphasizes the role of organizational culture and individual resilience in shaping employees' perceptions of technological change, suggesting that psychological outcomes are not solely determined by the technology itself but also by the broader social and managerial context.

Global trends indicate that the adoption of industrial robots is accelerating, particularly in Asia and Australia, with continued growth in Europe [16]. These developments highlight the importance of understanding how robotization affects employees

and of proactively preparing organizations and their workforce for ongoing technological change [13, 17].

Given the contradictory findings in the literature and the lack of in-depth qualitative analyses in the Central European context, there was a need to capture employees' subjective experiences. A qualitative approach enables a deeper understanding of how workers in different positions perceive robotization, what factors shape their experiences, what helps them cope with change, and where they identify the main risks. By exploring employees' narratives, researchers can identify both systemic issues and practical strategies that organizations can implement to support workers' well-being during technological transitions.

The objective of this study was to explore the impact of introducing robotic technologies on employees' mental and physical health across three Czech and one Slovak manufacturing company. The study draws on qualitative data obtained from interviews, complemented by workplace observations and the authors' professional expertise in occupational safety and health and workplace mental health.

The secondary objective is to identify risk and protective factors that may inform the development of preventive measures to minimize the psychological burden associated with the process of robotization.

2. METHODS

This qualitative study was conducted between 2023 and 2025 in four manufacturing companies where various types of robots (collaborative and industrial) had been implemented in the production process. The participating companies represented different manufacturing sectors, including automotive production, industrial assembly, and mechanical engineering, and varied in company size, degree of automation, and type of robotic technologies used. Additional characteristics of the participating companies are presented in Table 1.

Companies across the Czech Republic and Slovakia were invited to participate voluntarily. The inclusion of organizations representing different levels and forms of automation enabled the exploration of a broad range of employee experiences related to robotization. Within participating companies,

employees from production, supervisory, and technical positions were invited to participate. The participating organizations coordinated recruitment; however, participation was entirely voluntary, and employees could decline involvement without any consequences. Eligibility criteria required participants to have at least three months of experience working in a robotized work environment.

The final number of participating companies, interviews, and employees reflected not only the purposive sampling strategy but also several practical considerations inherent to workplace-based qualitative research. These included the substantial time required to conduct in-depth interviews, the research team's time and personnel capacity, and the need to minimize disruption to production processes, as interviews were conducted during working hours. Consequently, participant recruitment and interview scheduling were coordinated closely with company management to ensure employees could participate without adversely affecting operational activities.

In total, 27 semi-structured interviews were conducted with employees (15 women and 12 men/14 production operators, 6 shift coordinators (supervisors), and 7 technical staff members). In addition, 19 unstructured interviews were conducted with company managers (3 women and 16 men). The age of participants ranged from 20 to 61 years.

Before data collection, all respondents signed informed consent. The study was conducted in accordance with ethical standards for research involving human subjects. To ensure anonymity, participants'

names and other identifying information were removed from the transcripts, and all data were stored securely and accessible only to the research team. Given the non-interventional nature of the study, voluntary participation, and the anonymous collection of data, formal approval by an institutional ethics committee was not required.

Interviews were conducted directly at workplaces during regular working hours to ensure participants' convenience and contextual relevance. Semi-structured interviews with employees followed an interview guide covering topics such as perceived changes in work tasks, physical and mental health, stress, adaptation to robotization, cognitive demands, social relations, training experiences, and organizational support. The interview guide is provided in the Supplementary Materials. Unstructured interviews with managers focused on implementation processes, organizational challenges, employee adaptation, and management experiences related to robotization.

For organizational and confidentiality reasons, participating did companies was not permit audio recording to due of concerns about protecting internal processes and employee privacy.

Two trained researchers conducted each interview. One researcher facilitated the interview by asking the predefined questions, while the second researcher was solely responsible for contemporaneous note-taking. Detailed written notes were recorded throughout the interviews to capture participants' responses as accurately and comprehensively as possible, thereby enhancing the completeness and reliability of the qualitative data. Clarifying questions

Table 1. Characteristics of participating companies.

	Country	Manufacturing Sector	Company Size	Type of Robotization	Employee Interviews	Manager Interviews
1	Czech Republic	Automotive Industry	Large (~600)	Collaborative Robots	8	3
2	Czech Republic	Industrial Assembly	Medium (~160)	Industrial Robots	6	4
3	Slovak Republic	Mechanical Engineering	Large (~900)	Industrial and Collaborative Robots	10	6
4	Czech Republic	Automotive Industry	Large (~2000)	Industrial Robots and Automated Production Lines	3	6

were used during interviews when necessary to verify understanding, and interview notes were reviewed and supplemented immediately after data collection to improve completeness and accuracy.

Each interview lasted between 30 and 50 minutes. Variation in interview length reflected differences in participants' job roles and individual communication styles. Unstructured interviews with managers were generally longer because they addressed broader organizational issues and implementation processes.

The interviews were complemented by direct observations of the work environment and employees' interactions with technologies, conducted by members of the research team during site visits.

Data were analyzed manually using thematic analysis. Written interview records were reviewed repeatedly to ensure familiarity with the data and to identify meaningful units of text. In the first stage, open coding was applied to generate preliminary codes reflecting the key concepts, perceptions, and experiences reported by participants. In the subsequent stage, axial coding was used to examine relationships among codes and to organize them into broader conceptual categories. The initial open coding produced a wide range of preliminary codes, which were then refined, merged, and structured into higher-order categories during the axial coding process.

To strengthen analytical rigor, three members of the research team were involved in the coding process. Individual interviews were initially coded by one researcher and subsequently reviewed by another team member. Coding discrepancies were discussed in research team meetings until consensus was reached. Based on these discussions, the coding categories were further refined and applied consistently across the dataset. Through this iterative process of comparison, discussion, consensus-building, and refinement, the codes were grouped into nine final thematic areas representing the key dimensions of employees' experiences with robotization. An example of the coding structure and code development process is provided in Supplementary Material S2.

The multidisciplinary research team comprised psychologists, a psychiatrist, an occupational physician, and occupational health professionals with expertise in workplace mental health and occupational safety and health. The researchers acknowledged

that their professional backgrounds could influence both data collection and interpretation. To minimize individual bias, coding decisions and thematic interpretations were regularly discussed within the research team, and the findings were developed through consensus among researchers with different professional perspectives.

Formal approval by an institutional ethics committee was not required for this study, as it involved qualitative interviews with adult employees conducted as part of workplace research and did not include any medical interventions, the collection of biological samples, or the processing of sensitive personal data beyond participants' voluntary responses. Participation was entirely voluntary; all participants provided informed consent before the interviews, and anonymity and confidentiality were maintained throughout the study. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki.

3. RESULTS

The analysis of the interviews identified nine thematic areas: general perceptions, change, fear, problems, support (training and ongoing assistance), social relations and communication, employee characteristics, physical health, and mental health (including stress, fatigue, cognitive functions, and substance use). To provide greater clarity, the results are presented thematically, with each section summarizing the key findings and supported by selected quotations from employees and managers. These are coded according to the company and participant number, for example, C1P1 refers to Company 1, Participant 1.

3.1. General Perceptions

Employees in all four companies generally perceived robotization positively, particularly because it reduced physical strain, simplified certain tasks, and limited exposure to monotonous, hazardous, or physically demanding environments. Some employees had expected that the introduction of robots would also lead to a reduction in the pace of work; however, the reality was different—performance

requirements remained high, or even increased. Despite this, employees reported that robots had become an integral part of the workplace and were generally accepted without significant resistance. As one respondent noted: “From my point of view, it is positive that the work is physically less demanding, and looking ahead, robotization is inevitable as it continues to expand in other forms.” (C3P6).

3.2. Change

The process of introducing robots was described as a period of significant adjustment. As one participant put it: “Nobody likes changes, and nobody initially approaches them positively, but in my experience, these changes have always been for the better.” (C1P5). Employees highlighted the need to adapt quickly to new workflows, learn new technical procedures, and reorient themselves to an altered work environment. The initial adaptation period was generally considered the most demanding phase, both mentally and organizationally. “During the first month, I would have dismantled the robots and taken them back to the office... but later it became a relief for me and especially for my team.” (C1P2). Managers noted that implementing new technologies often required parallel organizational changes, such as redefining job roles and responsibilities.

3.3. Fear

Although most employees did not express strong fears related to working with robots, some concerns were reported. These included worries about making mistakes and the fear of being unable to keep up with technological demands. At the same time, respondents emphasized that they did not perceive robots as a direct physical threat. They trusted the well-established safety measures, adhered to protective instructions, and often described their feelings more as a form of healthy respect rather than fear. As one operator noted: “Well, we have fairly strict safety rules in place. If I follow them, I’m not afraid.” (C3P1). Both employees and managers repeatedly emphasized confidence in established safety procedures, which appeared to reduce concerns regarding physical injury.

3.4. Problems

Technical malfunctions and operational issues were among the most stressful aspects of robotization, particularly during the initial phase of implementation. Employees and managers emphasized that many of the difficulties were not caused by the robots themselves, but rather by inconsistent or unsuitable input materials. In several cases, participants attributed these operational problems to inconsistencies in the characteristics of input components, such as clips and other parts used in automotive manufacturing. Although these components generally complied with manufacturing specifications, even minor variations in their dimensions, shape, or positioning could interfere with the reliable operation of automated production processes. As one employee explained: “The robot itself is quite trouble-free. But the problem is with the bulk material... in my opinion, that is the worst part of robotization.” (C2P5). Another recurring concern was the absence of immediate technical support. One operator noted: “The worst thing is that on the second shift we don’t have a technologist. Some things just cannot be fixed by us; it has to be someone who really understands it.” (C4P2). Frequent troubleshooting increased stress, especially for technical staff who bore responsibility for ensuring continuous operation.

3.5. Support

Statements from both employees and managers indicated that high-quality training, accessible technical support, and a clearly defined *hierarchy* for assistance played a key role in the implementation and use of robots. As one participant noted: “There is technical support, we have manuals, and we have supervisors who are directly assigned to this, and we can communicate with them.” (C4P5). While training was considered important for all employees, it proved particularly demanding for technical staff, who had to master complex problem-solving skills, advanced programming, and the ability to manage unexpected malfunctions under time pressure. Supportive measures also included regular team meetings and informal sharing of experiences.

3.6. Social Relations and Communication

Robotization influenced team dynamics in several ways. In many workplaces, it fostered closer cooperation among operators, technicians, and managers, as employees needed to coordinate more frequently and share experiences to address emerging challenges. Open communication and informal knowledge exchange were described as essential for successful adaptation. As one employee explained: “I go to my colleagues who work before me in the process, and we solve problems together; now we are dependent on each other, so we communicate much more.” (C3P2). Interestingly, in some teams, acceptance of the technology was further supported by personifying the robots—for example, by giving them names or even “talking” to them. This practice reduced distance, lightened the atmosphere, and helped employees cope with the stress of working alongside automated systems.

3.7. Employee Characteristics

The characteristics of employees that proved important for managing work with robots varied by position type. For operators, formal qualifications were not decisive; rather, a positive attitude toward technology, motivation to work, manual skills, and the ability to learn new tasks were key. “I still believe that a person must want to do it, must have the will to learn; in my opinion, this work is for everyone.” (C4P2). For technical staff, crucial factors included technical competencies, the ability to solve problems under time pressure, and willingness to work in a three-shift system. “Certainly, they should understand programming to know what they are dealing with... And they must not act rashly, because a robot is a robot... Sometimes it is better to do it manually; there is no need to overthink it, one should act with deliberation and calm.” (C4P5). In practice, managers emphasized that one of the biggest challenges was not employees’ direct work with robots, but rather the recruitment and retention of qualified technical staff.

3.8. Physical Health

Employees consistently reported that robots helped reduce physical workload, limited repetitive strain, and minimized exposure to hazardous environments. Several respondents highlighted that this contributed to lower fatigue and fewer musculoskeletal complaints. In the words of one respondent: “When I started, everyone worked alone at their tables and did everything by hand. If it had always been done like that, I wouldn’t still be here. My hands were raw and bleeding. Robotization was a huge relief.” (C2P3). Conversely, in some workplaces, employees raised concerns about elevated noise levels and strong vibrations, which negatively affected employees’ overall psychological well-being during work. One technician noted: “...the noise and vibrations are enormous. It lasts all day. At the back, where you were, it’s fine, but here at the front, where the girls are, you have to shout and compete with the noise, and it’s psychologically exhausting. That noise wasn’t there before; it used to be much calmer.” (C2P5). It should be noted that these observations were based on employees’ subjective perceptions and were not verified through objective environmental measurements.

3.9. Mental Health

The greatest psychological demands were observed in the initial phases of robotization, when work processes were being reorganized, and employees had to rapidly adapt to the new system, acquire technical skills, and orient themselves in a changed work environment. As one employee described, “At first, it was a bit stressful until you understood what the robot was going to do.” (C4P8). Over time, however, employees reported that the situation improved as they became more familiar with the robots and the new workflows. Most stress was then experienced only in relation to technical problems, such as issues with input materials or malfunctions, rather than routine operations. One respondent noted, “...then you start getting frustrated, or when it happens near the end of the shift (those problems with the clips), it annoys you, and it affects your mental state.” (C1P5). It should also

be noted that increased noise and vibration levels in some workplaces continued to affect employees' psychological well-being negatively.

The mental health effects of robotization varied by job role. In most cases, operators did not perceive working with robots as particularly mentally or physically demanding, especially when they had received adequate training and had access to technical support. As one operator stated: "A person is not tired and can function even outside of work, at home." (C4P7).

Technical staff described increased stress related to their responsibility for the proper functioning of technologies, constant problem-solving, higher demands on learning, and limited opportunities for substitution by colleagues. "Like any job, working with robots can sometimes be stressful. It creates a certain stress in the worker—in my case, when something doesn't go well, when you can't set something up, or when the robot doesn't perform 100 percent as expected. From a mental perspective, one has to carefully think about what and how to do, which makes it more demanding." (C4P3). Although managers did not explicitly mention experiencing stress, their accounts indicated substantial psychological demands associated with coordinating robotized production processes.

They were responsible for coordinating and supervising the entire robotized production process, ensuring smooth operation, troubleshooting technical issues, managing staff schedules, and maintaining safety standards. In addition, they had to mediate between operators and technical staff, make rapid decisions under pressure, and respond to unexpected disruptions, all of which contributed to a high level of mental workload. Managers also faced considerable challenges in securing qualified employees, particularly technical staff, as there is no specialized education for these roles, and highly skilled individuals are often unwilling to work in a three-shift system.

The cognitive demands of work with robots largely stem from the nature of the tasks themselves. Operators particularly emphasized that working with robots required sustained attention and constant monitoring – a requirement that existed even before robotization. With robots, however, they now have

to closely observe the machines' processes, tracking both what the robot has completed and what remains to be done, which increases the level of attention required. "You have to be attentive, you need to watch whether the robot is stuck... it requires greater attention, to observe... to notice whether the robot is moving." (C1P3). Several respondents noted that the novelty of the robotized system initially increased cognitive load, as they had to remember new procedures, sequences, and safety protocols. Over time, as routines became established, cognitive demands decreased. Still, attention remained essential, especially during problem-solving tasks or when managing multiple processes simultaneously, which was particularly demanding for technical staff. The need to quickly detect and respond to errors or malfunctions placed high demands on concentration and working memory. As one employee reflected on the increased cognitive demands of working with robots and how they manifest beyond the workplace: "I think so, ever since I started working with robots, even at home when I do something, I find myself concentrating more, thinking about whether I'm doing it the right way or another way. It's like the focus I learned at work has carried over, and I find myself thinking more about everything I do." (C3P3).

Employees did not report any negative impact on sleep or the development of addictive behaviors. "For me, it's more about the three-shift system and the resulting irregular sleep. I don't think it's really about the robots, but rather about fatigue." (C1P1).

The implementation of robotic systems can mitigate certain occupational stressors, such as physical demands and exposure to hazardous environments. However, it may also introduce new risks, including technical malfunctions, elevated noise and vibration levels, and increased cognitive demands, which necessitate careful monitoring and proactive management to safeguard employee well-being.

Taken together, these nine thematic areas—ranging from general perceptions and adaptation to changes in physical and mental health—highlight the complex and multifaceted impacts of robotization on employees, encompassing both benefits and potential challenges that require ongoing attention.

4. DISCUSSION

The present qualitative study provides novel insights into the subjective experiences of employees and managers in Czech and Slovak manufacturing companies that have implemented industrial and collaborative robots. An analysis of interviews across four companies and multiple job positions highlights both the opportunities and challenges that accompany robotization in the workplace. The findings show that employees' perceptions and impacts of robotization are complex and significantly influenced by job type, the quality of organizational support, and the phase of adaptation. Altogether, these findings highlight that the perception and impact of robotization are dynamic and context-dependent, rather than uniform across all employees.

Overall, employees perceived robotization positively, particularly for reducing physical workload and eliminating monotonous or hazardous tasks. This is consistent with international evidence demonstrating that robots can decrease occupational risks and musculoskeletal strain [2, 3]. However, concerns were raised about elevated noise and vibration levels in certain workplaces, which employees linked to psychological fatigue. While large-scale European analyses have not identified statistically significant effects of robotization on the physical work environment [18], engineering research confirms that industrial robots inherently generate mechanical vibrations [19], and empirical measurements from robotic systems in healthcare settings indicate that noise levels may approach or exceed occupational safety thresholds [20]. These findings suggest that although not all participants reported experiencing excessive noise and vibration, such stressors may arise in specific workplace settings and should therefore be considered when designing occupational health and preventive interventions. Future research should integrate employees' subjective perceptions with objective environmental measurements to provide a more comprehensive assessment of these potential exposures and their impact on workers' health.

This duality—physical relief but potential sensory or cognitive strain—illustrates the complex health impacts of robotization and confirms earlier calls

for more holistic assessments of Industry 4.0 technologies [3].

Adaptation to new workflows was perceived as the most demanding phase, especially in the early stages of implementation. This is in line with Körner et al. [7], who noted that initial periods of human-machine interaction tend to involve heightened stress due to uncertainty and steep learning requirements [7]. Employees in this study often acknowledged that although change was initially difficult, it was eventually viewed as beneficial, echoing broader organizational research showing that technological transitions can yield long-term improvements despite initial resistance [1, 9]. On the other hand, some respondents noted that performance demands remained high or even increased, which, in some cases, contradicted their expectations. Such findings align with prior research emphasizing that while robots may reduce manual strain, they do not automatically lead to a slower or less pressured work environment [7, 13].

Regarding mental health, the findings were more nuanced. While operators generally did not experience increased psychological stress and trusted safety measures, technical staff perceived working with robots as more demanding, mainly due to higher responsibility, the unpredictability of technical problems, and the pressure to learn new skills continuously. Although managers did not explicitly report experiencing significant stress, they described a range of psychologically demanding responsibilities, including managing organizational change, reorganizing work processes, retraining employees, implementing technological innovations, and maintaining responsibility for production outcomes. Studies [1, 7, 8] confirm that technological changes do not necessarily affect employees' mental health equally across different positions or roles.

The finding that technical staff and managers experienced greater psychological demands than production operators may be interpreted in light of the Job Demands-Resources (JD-R) model [21]. According to this framework, employee well-being is influenced by the balance between job demands and available job resources. In the present study, production operators generally performed more structured, predictable tasks and often had access

to established procedures and technical support when problems arose. In contrast, technical staff faced elevated job demands associated with troubleshooting, responsibility for production continuity, continuous learning, and rapid problem-solving during technical failures. Managers were exposed to substantial organizational demands, including coordinating technological change, supervising employees, managing staffing challenges, and maintaining production performance. Although access to resources such as training, technical support, collaborative teams, and clear communication appeared to mitigate some of these demands, the overall balance between demands and resources seemed less favorable for technical staff and managers than for production operators. These findings suggest that the psychological impact of robotization is shaped not only by the technology itself but also by the balance between job demands and organizational resources available to different occupational groups.

Technical malfunctions and operational issues were identified as some of the most stressful aspects of robotization. Importantly, employees reported that problems often arose not from the robots themselves, but from inconsistent input materials. This mirrors findings from previous studies that highlight the interconnectedness of technological and organizational factors in shaping employees' experiences [5, 8]. Employees reported frustration when immediate support was unavailable, illustrating the psychological burden of being solely responsible for resolving malfunctions under time pressure. Similar observations have been made in studies showing that perceived lack of control over technology is a significant driver of stress and reduced well-being [7, 8].

The findings suggest that adequate training and accessible technical support are crucial protective factors in mitigating stress and ensuring the successful integration of robots. Employees praised and emphasized the importance of proper onboarding, noting that hands-on experience on the job ultimately provided the most valuable learning. This aligns with prior research showing that training combined with participatory communication can enhance confidence and reduce psychological strain, thereby mitigating the potential mental health

challenges associated with robotization [3, 13]. For technical staff, however, training remained more intensive, requiring the development of programming and troubleshooting skills, highlighting the need for tailored programs that reflect the specific demands of different roles.

Furthermore, respondents emphasized the importance of clear hierarchies of assistance. These organizational practices resonate with the broader literature on the role of organizational culture in shaping employee well-being during technological change [14].

An interesting observation was that employees in several teams tended to personify the robots they worked with by assigning them human names and, in some cases, even "talking" to them. Although such practices have received limited attention in the occupational health and human-robot interaction literature, participants described them as contributing to a more familiar and friendly relationship with the technology, reducing stress, and facilitating its acceptance in the workplace. By attributing human-like characteristics to robotic systems, employees may reduce the psychological distance between themselves and the technology, increase familiarity, and perceive the work environment as more approachable and less threatening. In this context, personification may represent an adaptive coping mechanism that supports successful human-robot collaboration, facilitates acceptance of technological change, and promotes the social integration of robotic systems into everyday work practices.

One notable finding is that employees expressed very little fear of physical harm from robots. Instead, they reported strong trust in established safety protocols and emphasized adherence to protective measures. This supports previous recommendations that well-designed safety standards are critical to preventing accidents and alleviating anxiety [22]. Rather than fear, employees described a form of "healthy respect" toward robots, which may serve as a psychological coping mechanism that balances awareness of potential risks with confidence in protective systems. This contrasts with studies in other contexts, such as China, where rapid robot adoption has been linked to heightened job insecurity and health concerns [10]. The respondents in this study

appeared more reassured, possibly due to structured safety procedures and closer integration of operators in the robotization process.

Cognitive demands were highlighted across several positions, with operators and technicians reporting the need for sustained attention, rapid error detection, and adherence to new procedures. This aligns with experimental evidence showing that human-robot collaboration can elevate perceived cognitive load, particularly in the early stages of adaptation [11]. An intriguing finding of the present study was that several employees reported heightened attentional focus extending beyond the workplace into their personal lives. This observation may reflect two distinct, but not mutually exclusive, mechanisms. On the one hand, it may indicate a positive cognitive adaptation, whereby working with robotic systems enhances sustained attention, vigilance, and attentional control.

On the other hand, it may represent a spillover of work-related cognitive demands into non-work domains, suggesting incomplete psychological detachment from work and a persistent cognitive load beyond working hours. Owing to the qualitative design of the present study, it was not possible to determine which of these mechanisms predominated or whether both contributed to the reported experiences. Future research should further investigate this phenomenon using longitudinal designs and objective measures of cognitive workload, attentional performance, recovery processes, and work-life spillover to clarify its underlying mechanisms and potential implications for employee well-being.

Notably, employees did not report significant sleep disturbances or substance use linked directly to robotization. Instead, fatigue was primarily associated with shift work, confirming that contextual work-organization factors may outweigh technological factors in shaping certain health outcomes [23].

Robotization also influenced team dynamics. Respondents reported closer cooperation between operators, technicians, and managers, with communication becoming more frequent and problem-solving more collaborative. This observation is supported by Berkers et al. [5], who found that robotization reshapes work design in logistics

by fostering interdependence and new patterns of cooperation.

The presented study, together with previous research [13, 22], suggests several recommendations that may be highly beneficial when implementing robots within a company's structure. Support from supervisors, the availability of technical staff, and regular opportunities for feedback appear to be crucial factors in reducing psychological stress and increasing employees' willingness to embrace change. Other important protective factors include accessible technical support, thorough training, a clearly defined hierarchy for assistance, and a work environment that promotes collaboration. Some teams also employed informal strategies to reduce psychological distance, such as naming robots and "personalizing" technology, which facilitated faster acceptance of the systems.

5. STRENGTHS AND LIMITATIONS

A major strength of this study lies in its qualitative design, which enabled an in-depth exploration of employees' and managers' subjective experiences with robotization. By including participants from different occupational groups, including production operators, technical staff, supervisors, and managers, the study captured a broad range of perspectives on the organizational, psychological, and physical impacts of robotization. The inclusion of companies that differed in size, sector, and degree of automation further enhanced the diversity of experiences represented in the data.

Several limitations should be considered. The participating companies were recruited voluntarily, and recruitment within each organization was coordinated by company representatives, which may have introduced selection bias. Organizations with more positive experiences of robotization may have been more inclined to participate. Furthermore, only current employees were included in the study. Individuals who had left their jobs because of difficulties associated with robotization, such as unsuccessful adaptation, job displacement, or dissatisfaction, were not represented, creating a potential survivorship bias.

At the same time, this characteristic of the study provided an opportunity to explore the factors associated with long-term retention in robotized work environments. The findings offer insights into the individual, organizational, and workplace characteristics that may facilitate successful adaptation to robotization and influence which employees remain in these work settings over time. Although these observations should be interpreted cautiously, they represent an important complementary outcome of the study. They may inform future longitudinal research on workforce adaptation and retention in increasingly automated workplaces.

The relatively small sample size and the study's context-specific nature limit the direct transferability of the findings to other organizational settings. Nevertheless, the identified themes and underlying mechanisms provide valuable insights into employees' experiences of workplace robotization that may apply to organizations with similar technological and organizational characteristics. Thus, while the findings are not intended to support statistical generalization, they contribute to the conceptual understanding of employee adaptation to robotized work environments and may guide future research, policy, and practice.

Another limitation of this study is that the interviews were not audio-recorded due to organizational confidentiality requirements. Although the absence of verbatim transcripts may have limited the capture of certain nuances in participants' responses, this potential limitation was mitigated by conducting each interview with two trained members of the research team. While one researcher led the interview and asked the predefined questions, the second researcher was dedicated exclusively to contemporaneous note-taking. The notes were reviewed immediately after each interview to ensure completeness and accuracy, thereby enhancing the credibility of the qualitative data collected.

In addition, several participants reported experiencing excessive workplace noise and vibrations during the interviews. However, these observations were based exclusively on employees' subjective perceptions. They were not verified by objective environmental measurements, as assessing physical workplace exposures was beyond the scope and primary objectives of the present study. Consequently, these findings should

be interpreted as exploratory observations rather than evidence of actual exposure levels.

Despite these limitations, the study provides valuable insights into employees' experiences of robotization and identifies organizational factors that may support successful adaptation to technological change.

6. CONCLUSION

This study explored the impact of introducing robotic technologies on employee health in manufacturing companies in the Czech Republic and Slovakia, and identified key risk and protective factors to inform the development of preventive measures to reduce the psychological burden associated with robotization.

The results suggest that robotization can have positive effects on employees' physical health and, under certain conditions, pose no significant risks to their psychological well-being. A key factor is the quality of the implementation process—comprehensive training, accessible technical support, a clearly defined hierarchy of assistance, and effective team communication. It appears crucial to monitor the mental workload of technical and managerial staff, to create new job profiles (e.g., a mechatronics specialist serving as an intermediary between operators and technicians), and to foster a team culture in which mutual support and knowledge sharing are standard practice.

The results underscore that the impact of robotization on health and well-being is shaped not only by the technology itself but also by organizational support, team dynamics, and individual characteristics. By addressing these factors proactively, companies can harness the benefits of robotization while minimizing its risks.

Further research is necessary to gain a deeper understanding of the complex impacts of robotization and to identify strategies to minimize its risks while fostering a healthy work environment.

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APPENDIX

SUPPLEMENTARY MATERIAL S1: INTERVIEW GUIDE

1. Demographic and Occupational Characteristics
 - Age
 - Education
 - Current job position and work tasks
 - Length of employment in the company and current position
 - Previous experience with robotic technologies
2. General Perceptions of Robotization
 - How do you perceive the introduction of robots into your work?
 - What advantages and disadvantages do you associate with robotization?
 - Has working with robots affected your health or quality of life?
3. Changes Associated with Robotization
 - How has your work changed since the introduction of robots?
 - How has robotization affected your personal life?
 - What other changes have you noticed following robot implementation?
4. Employee Characteristics and Suitability for Robotized Work
 - What personal characteristics or skills are important for working with robots?
 - Are there groups of people who may find working with robots more difficult?
5. Training and Organizational Support
 - How was your training for working with robots organized?
 - What aspects of the training were useful or insufficient?
 - What forms of support are available when problems occur?
 - What additional support would help you perform your work more effectively?
6. Technical Problems and Error Management
 - What problems most commonly occur when working with robots?
 - How do you usually respond when technical problems arise?
 - How stressful are such situations?
 - To what extent can you rely on colleagues and supervisors when difficulties occur?
7. Cognitive Demands
 - Has working with robots influenced your attention or concentration?
 - Has it influenced your memory or information processing?
 - How mentally demanding do you find robotized work?
8. Mental Health and Well-being
 - How stressful is working with robots compared with previous work methods?
 - How does robotized work affect fatigue and recovery during shifts?
 - Has robotization influenced your sleep, well-being, or psychological health?
 - Do you have any concerns about safety, accidents, or health risks associated with robots?
9. Physical Health
 - Has robotization influenced your physical workload?
 - Have you noticed any positive or negative effects on your physical health?
10. Social Relationships and Communication
 - Has robotization changed communication with colleagues?
 - Has it changed communication with supervisors?
 - How important is cooperation in robotized work environments?
11. Lifestyle and Habits
 - Has working with robots influenced any lifestyle habits or behaviors, such as alcohol consumption, smoking, social media use, gaming, or other activities?
12. Overall Evaluation
 - Looking back, how would you summarize your experience with robotization?
 - Is there anything you would like to add or revise regarding your previous answers?

Table S2. Example of coding procedure.

Raw data extract:	Initial code	Category	Final theme
“It is a relief, but at the beginning it was terrible until everything was set up properly.”	Positive perception of robotization	General evaluation of robotization	General perceptions
“Nobody likes changes at first, but later it became a relief for me and my team.”	Adaptation to new technology	Organizational change	Change
“If I follow the safety rules, I am not afraid.”	Trust in safety measures	Safety perception	Fear
“The clips are not always the same and the robot gets stuck.”	Technical malfunction	Operational difficulties	Problems
“The worst thing is when technical support is not available.”	Lack of support during problems	Operational difficulties	Problems
“We have supervisors, manuals, and technical support available.”	Available assistance	Organizational support	Support
“We help each other and solve problems together.”	Knowledge sharing	Team cooperation	Social relations and communication
“We call the robot Lojza.”	Personification of robots	Informal coping strategy	Social relations and communication
“A person must want to learn new things.”	Motivation to learn	Employee readiness	Employee characteristics
“Robotization was a huge relief for my hands.”	Reduced physical strain	Lower physical workload	Physical health
“The noise and vibrations are enormous.”	Noise and vibration exposure	Workplace environmental stressors	Physical health
“I have to concentrate more and constantly monitor the robot.”	Increased attention	Cognitive demands	Mental health
“When something does not work, it creates stress.”	Stress during troubleshooting	Psychological demands	Mental health
“I do not think robotization affects my sleep.”	No sleep-related problems	Sleep and recovery	Mental health